

# An Inverse Riccati Construction of Second-Order Linear ODEs from Degree-2 Algebraic Extensions

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## Abstract

We present an explicit two-pole construction of a second-order linear ODE of the form  $y'' + r(z)y = 0$  whose solutions are Liouvillian and arise from a logarithmic derivative  $s(z) \in \mathbb{C}(z, \sqrt{R})$  of algebraic degree exactly 2 over  $\mathbb{C}(z)$ . The construction places two poles at  $z = 0$  and  $z = 1$  with equal residues  $1/4$ , yielding the potential  $r(z) = \frac{3}{16z^2(z-1)^2}$  and solutions expressible in terms of radicals and inverse hyperbolic functions. We verify the Riccati equation by direct computation, prove that the logarithmic derivative has algebraic degree 2 via its minimal polynomial, and establish linear independence of the two fundamental solutions via their nonconstant ratio and constant Wronskian. An outline of an  $n$ -pole generalization with a complete irrational cancellation identity is included. We have not found this exact inverse two-pole construction in the standard references consulted.

## 1 Introduction

The Kovacic algorithm [1] provides a systematic method for determining whether a second-order linear ODE of the form

$$y'' + r(z)y = 0 \tag{1}$$

has Liouvillian solutions, i.e., solutions expressible in terms of exponentials, logarithms, and algebraic functions. The algorithm operates by examining the poles of the coefficient function  $r(z)$  and classifying the local behavior at each singular point into one of four cases, each of which constrains the possible algebraic degree of the logarithmic derivative of a solution.

A Liouvillian solution of (1) can be written as  $y_1 = \exp(\int s(z) dz)$ , where  $s(z) \in \mathbb{C}(z, \sqrt{R})$  satisfies the Riccati equation

$$s' + s^2 + r(z) = 0. \tag{2}$$

The algebraic degree of  $s(z)$  over  $\mathbb{C}(z)$  determines the complexity of the solution. Kovacic's algorithm searches for  $s(z)$  of degree 1, 2, 4, or 8 over  $\mathbb{C}(z)$ . In this paper we focus on degree-2 solutions, where  $s(z) \in \mathbb{C}(z, \sqrt{R})$  for some square-free  $R \in \mathbb{C}(z)$  with  $\sqrt{R} \notin \mathbb{C}(z)$ .

We present a construction that starts from the desired algebraic structure—a quadratic extension  $\mathbb{C}(z, \sqrt{R})/\mathbb{C}(z)$ —and derives the ODE coefficient  $r(z)$  from the Riccati equation. This is the reverse of the usual approach, which starts from  $r(z)$  and attempts to find  $s(z)$ . Our construction naturally produces ODEs with two poles, in contrast to many of the single-pole or single-point examples we encountered in the references consulted within Kovacic's framework.

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## 2 The Two-Pole Ansatz

We seek a logarithmic derivative of the form

$$s(z) = A(z) + B(z)\sqrt{R(z)}, \quad A, B, R \in \mathbb{C}(z), \quad \sqrt{R} \notin \mathbb{C}(z), \quad (3)$$

where  $s(z)$  lies in the quadratic extension  $\mathbb{C}(z, \sqrt{R})$  of  $\mathbb{C}(z)$ . The Riccati equation implies  $r(z) = -(s' + s^2)$ .

We place two poles at  $z = 0$  and  $z = 1$  with equal residues  $1/4$ . This gives

$$A(z) = \frac{2z-1}{4z(z-1)} = \frac{1}{4z} + \frac{1}{4(z-1)}, \quad (4)$$

$$R(z) = \frac{1}{z(z-1)}, \quad (5)$$

$$B = c = \frac{1}{2}. \quad (6)$$

For the explicit two-pole construction we take  $B = c = 1/2$ , a constant. In this constant- $B$  case, the irrational part of  $s' + s^2$  vanishes when

$$\frac{R'}{2} + 2AR = 0, \quad (7)$$

which is verified by direct computation in Theorem 1. The resulting potential depends on  $c^2$ ; the choice  $c = 1/2$  gives the simple potential used below. The nonconstant- $B$  cancellation condition and its  $n$ -pole parametrization are developed in Section 7. This yields the two branches

$$s_{\pm}(z) = \frac{2z-1}{4z(z-1)} \pm \frac{1}{2\sqrt{z(z-1)}}. \quad (8)$$

The two poles of  $A$  are located at  $z = 0$  and  $z = 1$ , each with residue  $1/4$ ; the branch cut for  $\sqrt{z(z-1)}$  is taken along  $[0, 1]$ .

### 2.1 Field-Extension Tower

$$\begin{array}{c} \mathbb{C}(z, \sqrt{R}) \quad [\text{degree } 2] \\ \uparrow \sqrt{R} \notin \mathbb{C}(z) \\ \mathbb{C}(z) \quad [\text{rational functions}] \\ \text{Quadratic extension for } s(z). \end{array}$$

## 3 The Construction

### 3.1 Choosing the Poles and Computing $A$ , $B$ , $R$ , $s$

We choose poles at  $z_1 = 0$  and  $z_2 = 1$  with residues  $a_1 = a_2 = 1/4$  (corresponding to  $n_1 = n_2 = 1$ , both odd). The partial-fraction decomposition of  $A$  is

$$A(z) = \frac{2z-1}{4z(z-1)} = \frac{1}{4z} + \frac{1}{4(z-1)}.$$

The radicand  $R$  encodes the irrational part of the logarithmic derivative:

$$R(z) = \frac{1}{z(z-1)}.$$

With  $B = c = 1/2$  (the simplest subfamily,  $f(z) = 1$ ), the two branches of the logarithmic derivative are

$$s_{\pm}(z) = \frac{2z-1}{4z(z-1)} \pm \frac{1}{2\sqrt{z(z-1)}}.$$

## 4 The ODE and Its Solutions

### 4.1 The Potential $r(z)$

From the Riccati equation,  $r(z) = -(s' + s^2)$ . We first compute the rational part  $A' + A^2 + B^2R$ . The irrational part is shown below to vanish for this choice of  $A$  and  $R$ :

$$A' = -\frac{1}{4z^2} - \frac{1}{4(z-1)^2}, \quad (9)$$

$$A^2 = \frac{1}{16z^2} + \frac{1}{8z(z-1)} + \frac{1}{16(z-1)^2}, \quad (10)$$

$$B^2R = \frac{1}{4z(z-1)}. \quad (11)$$

Summing these three contributions,

$$A' + A^2 + B^2R = -\frac{3}{16z^2} - \frac{3}{16(z-1)^2} + \frac{3}{8z(z-1)} = -\frac{3}{16z^2(z-1)^2}. \quad (12)$$

Therefore

$$r(z) = -(A' + A^2 + B^2R) = \frac{3}{16z^2(z-1)^2}. \quad (13)$$

The irrational parts cancel because  $R'/2 + 2AR = 0$  for our specific  $A$  and  $R$ , as verified explicitly in Theorem 1. Thus both branches yield the same  $r(z) \in \mathbb{C}(z)$ .

### 4.2 Integrating $s$ to Obtain $y_1$

The first solution is

$$y_1 = \exp\left(\int s_+ dz\right). \quad (14)$$

We split the integral:

$$\int s_+ dz = \int \frac{2z-1}{4z(z-1)} dz + \int \frac{1}{2\sqrt{z(z-1)}} dz. \quad (15)$$

The first integral is elementary, and the second is expressed using arcosh:

$$\int s_+ dz = \frac{1}{4} \log(z(z-1)) + \frac{1}{2} \operatorname{arcosh}(2z-1). \quad (16)$$

Using the complex identity  $\operatorname{arcosh}(2z-1) = \log(2z-1 + 2\sqrt{z(z-1)})$ , valid after choosing compatible branches, we obtain  $\int s_+ dz = \frac{1}{4} \log(z(z-1)) + \frac{1}{2} \log(2z-1 + 2\sqrt{z(z-1)})$ . Exponentiating gives

$$y_1 = [z(z-1)]^{1/4} [2z-1 + 2\sqrt{z(z-1)}]^{1/2}. \quad (17)$$

### 4.3 The Second Solution $y_2$ from the Conjugate

The second solution arises from the conjugate logarithmic derivative

$$s_2 = A - B\sqrt{R} \quad (18)$$

obtained by applying the nontrivial automorphism of the quadratic extension  $\mathbb{C}(z, \sqrt{R})/\mathbb{C}(z)$  which sends  $\sqrt{R} \mapsto -\sqrt{R}$ . This yields

$$y_2 = [z(z-1)]^{1/4} [2z-1 - 2\sqrt{z(z-1)}]^{1/2}. \quad (19)$$

The automorphism preserves the Riccati equation because  $r(z) \in \mathbb{C}(z)$  is fixed, so  $s_2$  is also a root and  $y_2 = \exp(\int s_2 dz)$  is also a solution.

## 5 Proofs

**Theorem 1** (Riccati Verification). *Let  $A(z) = \frac{2z-1}{4z(z-1)}$ ,  $B = \frac{1}{2}$ ,  $R(z) = \frac{1}{z(z-1)}$ , and  $s_{\pm} = A \pm B\sqrt{R}$ . Define  $r(z) = -(A' + A^2 + B^2R)$ . Then*

$$r(z) = \frac{3}{16z^2(z-1)^2}$$

and both  $s_+$  and  $s_-$  satisfy the Riccati equation  $s' + s^2 + r(z) = 0$ .

*Proof.* The rational computation is given in (13). It remains to verify that the irrational part vanishes. For  $s_+ = A + B\sqrt{R}$ ,

$$s'_+ + s_+^2 = A' + A^2 + B^2R + \left( \frac{BR'}{2\sqrt{R}} + 2AB\sqrt{R} \right).$$

The irrational part is  $\frac{B}{\sqrt{R}} \left( \frac{R'}{2} + 2AR \right)$ . For our specific  $A$  and  $R$ ,

$$\frac{R'}{2} + 2AR = -\frac{2z-1}{2z^2(z-1)^2} + \frac{2(2z-1)}{4z(z-1)} \cdot \frac{1}{z(z-1)} = -\frac{2z-1}{2z^2(z-1)^2} + \frac{2z-1}{2z^2(z-1)^2} = 0.$$

Thus the irrational part vanishes, and  $s'_+ + s_+^2 = A' + A^2 + B^2R = -r(z)$ . The case  $s_- = A - B\sqrt{R}$  is identical with  $B$  replaced by  $-B$ ; since the irrational term is proportional to  $B$  and equals zero, both branches satisfy the Riccati equation.  $\square$

**Theorem 2** (Algebraic Degree 2). *The logarithmic derivative  $s = A + B\sqrt{R}$  has algebraic degree exactly 2 over  $\mathbb{C}(z)$ ; that is,  $s \notin \mathbb{C}(z)$  and  $s$  satisfies a degree-2 minimal polynomial over  $\mathbb{C}(z)$ .*

*Proof.* Since  $s = A + B\sqrt{R}$  with  $B \neq 0$ , the minimal polynomial is obtained by eliminating  $\sqrt{R}$ :  $(s - A)^2 = B^2R$ , or

$$s^2 - 2As + (A^2 - B^2R) = 0. \quad (20)$$

Substituting  $A = (2z-1)/(4z(z-1))$ ,  $B = 1/2$ , and  $R = 1/(z(z-1))$  gives:

$$s^2 - \frac{2z-1}{2z(z-1)}s + \frac{1}{16z^2(z-1)^2} = 0. \quad (21)$$

Here  $R = 1/(z(z-1))$  is not a square in  $\mathbb{C}(z)$ , since it has simple poles at  $z = 0$  and  $z = 1$ , while a square in  $\mathbb{C}(z)$  can only have even-order poles. Thus  $\sqrt{R} \notin \mathbb{C}(z)$ ; since  $B \neq 0$ ,  $s \notin \mathbb{C}(z)$ . The polynomial (21) has degree 2 in  $s$  with coefficients in  $\mathbb{C}(z)$ . Since  $s \notin \mathbb{C}(z)$ , it is irreducible over  $\mathbb{C}(z)$  and hence is the minimal polynomial.  $\square$

**Theorem 3** (Linear Independence). *The solutions  $y_1$  and  $y_2$  are linearly independent over  $\mathbb{C}$ .*

*Proof.* Set  $u = 2z - 1 + 2\sqrt{z(z-1)}$  and  $v = 2z - 1 - 2\sqrt{z(z-1)}$ . The identity  $(2z-1)^2 - 4z(z-1) = 1$  gives  $uv = 1$ . On any branch where  $y_1$  and  $y_2$  are defined, their ratio is

$$\frac{y_1}{y_2} = \left( \frac{u}{v} \right)^{1/2}.$$

Since  $v = 1/u$ , this ratio is  $\pm u$ , depending on branch choices. But  $u = 2z - 1 + 2\sqrt{z(z-1)}$  is a nonconstant algebraic function. Therefore  $y_1/y_2$  is not constant on any open domain where the branches are defined, and  $y_1, y_2$  are linearly independent.

Additionally, by Abel's identity, the Wronskian is constant because the ODE has no  $y'$  term. Direct evaluation gives  $W(y_1, y_2) = \pm 1$ , which is constant and nonzero on each branch.  $\square$

## 6 Computational Verification

The following SymPy session verifies the key identities computationally. This section is provided for reproducibility and is separate from the formal proofs above.

Listing 1: SymPy Verification Session

```
1 import sympy as sp
2 z = sp.Symbol('z')
3
4 # Define A, B, R
5 A = (2*z - 1) / (4*z*(z - 1))
6 B = sp.Rational(1, 2)
7 R = 1 / (z*(z - 1))
8
9 # Define s_+ and s_-
10 s_plus = A + B * sp.sqrt(R)
11 s_minus = A - B * sp.sqrt(R)
12
13 # Verify r(z) = -(A' + A^2 + B^2*R)
14 r = -(sp.diff(A, z) + A**2 + B**2*R)
15 r_simplified = sp.simplify(r)
16 print("r(z)=", r_simplified)
17 # Output: 3/(16*z**2*(z-1)**2) -- OK
18
19 # Verify Riccati: s' + s^2 + r = 0 for both branches
20 ric_plus = sp.simplify(sp.diff(s_plus, z) + s_plus**2 + r)
21 ric_minus = sp.simplify(sp.diff(s_minus, z) + s_minus**2 + r)
22 print("Riccati_+:", ric_plus) # 0 -- OK
23 print("Riccati_-:", ric_minus) # 0 -- OK
24
25 # Verify minimal polynomial
26 s = sp.Symbol('s')
27 minpoly = s**2 - 2*A*s + (A**2 - B**2*R)
28 print("Minimal_polynomial:", sp.simplify(minpoly))
29 # degree 2 in s -- OK
30
31 # Verify key identity: (2z-1)^2 - 4z*(z-1) = 1
32 identity = sp.expand((2*z - 1)**2 - 4*z*(z - 1))
33 print("(2z-1)^2-4z(z-1)=", identity) # 1 -- OK
34
35 # Verify solutions
36 y1 = (z*(z-1))**sp.Rational(1,4) * \
37     (2*z - 1 + 2*sp.sqrt(z*(z-1)))**sp.Rational(1,2)
38 y2 = (z*(z-1))**sp.Rational(1,4) * \
39     (2*z - 1 - 2*sp.sqrt(z*(z-1)))**sp.Rational(1,2)
40
41 # Check ODE: y'' + r*y = 0
42 ode_check_1 = sp.simplify(sp.diff(y1, z, 2) + r * y1)
43 ode_check_2 = sp.simplify(sp.diff(y2, z, 2) + r * y2)
44 print("ODE_check_y1:", sp.simplify(ode_check_1)) # 0 -- OK
45 print("ODE_check_y2:", sp.simplify(ode_check_2)) # 0 -- OK
46
47 # Wronskian
48 W = sp.simplify(y1 * sp.diff(y2, z) - sp.diff(y1, z) * y2)
49 print("Wronskian:", W)
50 # Branch-aware simplification gives W = +/- 1 (nonzero) -- OK
51
52 # Ratio y1/y2
```

```

53 ratio = sp.simplify(y1 / y2)
54 print("y1/y2=", ratio)
55 # Using uv=1, manual simplification gives y1/y2 = +/- u (nonconstant)
    -- OK

```

All computational checks confirm the theoretical results. The SymPy verification is consistent with the formal proofs in Section 5.

## 7 Outline of an $n$ -Pole Generalization

The construction extends formally to  $n$  poles at  $z_1, \dots, z_n$  with residues  $a_i = n_i/4$ , where each  $n_i$  is odd. The framework is

$$A(z) = \sum_{i=1}^n \frac{a_i}{z - z_i}, \quad a_i = \frac{n_i}{4}, \quad n_i \in \{1, 3, 5, \dots\} \quad (22)$$

$$\exp\left(-2 \int A dz\right) = \prod_{i=1}^n (z - z_i)^{-2a_i} = \prod_{i=1}^n (z - z_i)^{-n_i/2}. \quad (23)$$

The square root of  $R$  is then

$$\sqrt{R} = \frac{\prod_{i=1}^n (z - z_i)^{-n_i/2}}{f(z)}, \quad f(z) \in \mathbb{C}(z). \quad (24)$$

The radicand and coefficient  $B$  are

$$R = \frac{\prod_{i=1}^n (z - z_i)^{-n_i}}{f(z)^2} \quad (25)$$

$$B = cf(z). \quad (26)$$

For a general rational coefficient  $B(z)$ , the irrational part of  $s' + s^2$ , with  $s = A + B\sqrt{R}$ , vanishes precisely when

$$\frac{B'}{B} + \frac{R'}{2R} + 2A = 0. \quad (27)$$

Taking

$$B = cf(z), \quad R = \frac{\prod_{i=1}^n (z - z_i)^{-n_i}}{f(z)^2} \quad (28)$$

satisfies this identity identically, because

$$\frac{B'}{B} = \frac{f'}{f}, \quad \frac{R'}{2R} = -2A - \frac{f'}{f}, \quad (29)$$

where the second identity follows from logarithmic differentiation of  $R = \prod_i (z - z_i)^{-n_i} / f(z)^2$ . Therefore

$$\frac{B'}{B} + \frac{R'}{2R} + 2A = \frac{f'}{f} + \left(-2A - \frac{f'}{f}\right) + 2A = 0. \quad (30)$$

In the simplest subfamily  $f(z) = 1$ , one has  $B = c$ , a constant, recovering the two-pole example treated above. The ODE coefficient is  $r(z) = -(A' + A^2 + B^2 R)$ , which depends on  $c^2$  through the  $B^2 R$  term, yielding a one-parameter family of potentials indexed by  $c$ .

## 8 Comparison with Kovacic's Examples

Kovacic's original paper [1] and subsequent literature, such as [2, 3], study ODEs with a single irregular singular point or specific pole configurations arising from special-function theory, including Bessel, Airy, and hypergeometric examples. Our construction differs from examples we encountered in the consulted references in the following sense:

| Feature                | Examples encountered in consulted references                     | This construction                             |
|------------------------|------------------------------------------------------------------|-----------------------------------------------|
| Pole structure         | Single or multi-pole configurations                              | Two distinct poles at $z = 0$ and $z = 1$     |
| Explicit $r(z)$        | Varies, e.g., $\frac{1}{4z^2}$ , $\frac{-1+3z^2}{4z^2(z^2-1)^2}$ | $\frac{3}{16z^2(z-1)^2}$                      |
| Algebraic extension    | Examples such as $\mathbb{C}(z, \sqrt{az+b})$                    | $\mathbb{C}(z, \sqrt{1/(z(z-1))})$            |
| Construction direction | Given $r(z)$ , find $s(z)$                                       | Given the structure of $s(z)$ , derive $r(z)$ |
| Residue pattern        | Various                                                          | Equal residues $1/4$ at each pole             |

In the examples from Kovacic and the references we consulted [1, 2, 3, 4], we did not find this exact inverse two-pole construction. The construction is natural from the perspective of inverse Riccati problems but appears to be less common in the examples we consulted.

## 9 Conclusion

We have presented a systematic construction of second-order linear ODEs with degree-2 Liouvillian solutions, based on a two-pole ansatz for the logarithmic derivative  $s(z) = A(z) + B(z)\sqrt{R(z)}$ . The specific example with poles at  $z = 0$  and  $z = 1$  yields the ODE  $y'' + \frac{3}{16z^2(z-1)^2}y = 0$  with solutions

$$y_1 = [z(z-1)]^{1/4}[2z-1+2\sqrt{z(z-1)}]^{1/2}$$

$$y_2 = [z(z-1)]^{1/4}[2z-1-2\sqrt{z(z-1)}]^{1/2}$$

The construction was verified by: (i) direct computation of  $r(z)$  from  $-(A' + A^2 + B^2R)$ ; (ii) proof that  $s$  has algebraic degree exactly 2 via its minimal polynomial; (iii) proof of linear independence via the nonconstant ratio  $y_1/y_2$  and constant nonzero Wronskian  $W = \pm 1$ ; and (iv) supporting SymPy computational checks.

The  $n$ -pole generalization extends the framework to arbitrary numbers of poles with odd residue parameters, with the simplest subfamily  $f(z) = 1$  yielding constant  $B = c$ . Further exploration of nonconstant  $f(z)$  and the resulting ODE coefficients is a natural direction for future work.

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reviewed the mathematical claims, computations, citations, and conclusions, and accepts full responsibility for the manuscript.



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